FLOWMO

Flow to the Mode

Mode-Seeking Diffusion Autoencoders for Image Tokenization | 2025

Train a transformer diffusion autoencoder first to match the reconstruction distribution, then push its samples toward perceptually faithful modes.

tokenization · rectified flow · mode seeking

1 Problem

THE TOKENIZER BOTTLENECK

- Two-stage image generators depend on a compact tokenizer, whose rate-distortion-perception trade-off limits the whole system.
- Dominant tokenizers use CNNs, 2D-aligned codes, adversarial losses or distillation from another tokenizer.
- Earlier diffusion autoencoders had not reached state-of-the-art ImageNet reconstruction.

2 Core insight

MATCH, THEN SEEK

- The reconstruction distribution p(x|c) is multimodal because discrete code c discards information.
- For perceptual reconstruction, matching every possible mode is less useful than sampling a mode close to the original image.
- FlowMo separates mode-matching pre-training from mode-seeking post-training.

3 Architecture

- A symmetric MM-DiT encoder/decoder maps an image to a 1D latent sequence and back through conditional rectified flow.
- Lookup-free quantization binarizes latent tokens; no convolution, GAN loss or auxiliary tokenizer is required.
- Stage 2 can train MaskGIT over the learned discrete token sequence.

4 Objectives

KEY EQUATIONS

- `x_t = t z + (1-t)x; L_flow = E ||x-z-d(x_t,c,t)||^2`
- Stage 1A adds perceptual, commitment and entropy losses to end-to-end flow matching.
- Stage 1B freezes the encoder and backpropagates a perceptual loss through an n-step ODE sample; a shifted sampler biases useful modes.

5 Evidence

STATE OF THE ART

- At 0.070 BPP, FlowMo-Lo reaches rFID 0.95 and PSNR 22.07; OpenMAGVIT-V2 gives 1.17 / 21.63.
- At 0.219 BPP, FlowMo-Hi reaches rFID 0.56 and PSNR 24.93, edging LlamaGen-32.
- Post-training improves Lo from 1.10 to 0.95 rFID and Hi from 0.73 to 0.56; generation matches strong traditional tokenizers.

6 Meaning & limits

TAKEAWAY

- A diffusion decoder can be a top-tier discrete tokenizer without conventional visual inductive biases.
- The ablations show small patches, perceptual loss, thick-tailed noise, shifted steps and guidance are all critical.
- Main limitation: decoding needs multiple forward passes (25 in the paper), so inference is slower.